

JDBC Programs

Reg no:24b01a12a6

1. Write a Java program to establish a connection with a MySQL database using JDBC and display a success message.

```
import java.sql.*;

public class Q1 {

    public static void main(String args[]) {

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/studentdb",
                "root",
                "root"
            );

            System.out.println("Connection Successful");

            con.close();

        } catch (Exception e) {

            System.out.println(e);

        }

    }

}
```

2. Write a Java program to display the database name, driver name, driver version, and URL using DatabaseMetaData.

```
import java.sql.*;

public class Q2 {

    public static void main(String args[]) {

        try {
```

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```
Class.forName("com.mysql.cj.jdbc.Driver");
```

```
Connection con = DriverManager.getConnection(
    "jdbc:mysql://localhost:3306/studentdb",
    "root",
    "root"
);
```

```
DatabaseMetaData db = con.getMetaData();
```

```
System.out.println("Database : " + db.getDatabaseProductName());
```

```
System.out.println("Driver : " + db.getDriverName());
```

```
System.out.println("Version : " + db.getDriverVersion());
```

```
System.out.println("URL : " + db.getURL());
```

```
con.close();
```

```
} catch (Exception e) {
```

```
    System.out.println(e);
```

```
}
```

```
}
```

```
}
```

3. Write a Java program to create a table named Student with appropriate fields using JDBC.

```
import java.sql.*;
```

```
public class Q3 {
```

```
    public static void main(String args[]) {
```

```
        try {
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
```

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```
Connection con = DriverManager.getConnection(
    "jdbc:mysql://localhost:3306/studentdb",
    "root",
    "root"
);
```

```
Statement st = con.createStatement();
```

```
String sql = "create table Student(id int,name varchar(30),marks int)";
```

```
st.executeUpdate(sql);
```

```
System.out.println("Table Created");
```

```
con.close();
```

```
} catch (Exception e) {
```

```
    System.out.println(e);
```

```
}
```

```
}
```

```
}
```

4. Write a Java program to insert a single record into the Student table using the Statement interface.

```
import java.sql.*;
```

```
public class Q4 {
```

```
    public static void main(String args[]) {
```

```
        try {
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            Connection con = DriverManager.getConnection(
```

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```
"jdbc:mysql://localhost:3306/studentdb",  
"root",  
"root"  
);
```

```
Statement st = con.createStatement();
```

```
st.executeUpdate("insert into Student values(101,'Ravi',85)");
```

```
System.out.println("Record Inserted");
```

```
con.close();
```

```
} catch (Exception e) {
```

```
System.out.println(e);
```

```
}
```

```
}
```

```
}
```

5. Write a Java program to retrieve and display all records from the Student table using ResultSet.

```
import java.sql.*;
```

```
public class Q5 {
```

```
public static void main(String args[]) {
```

```
try {
```

```
Class.forName("com.mysql.cj.jdbc.Driver");
```

```
Connection con = DriverManager.getConnection(  
    "jdbc:mysql://localhost:3306/studentdb",  
    "root",  
    "root"
```

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```
);
```

```
Statement st = con.createStatement();
```

```
ResultSet rs = st.executeQuery("select * from Student");
```

```
while (rs.next()) {
```

```
    System.out.println(rs.getInt(1) + " " +
```

```
        rs.getString(2) + " " +
```

```
        rs.getInt(3));
```

```
}
```

```
con.close();
```

```
} catch (Exception e) {
```

```
    System.out.println(e);
```

```
}
```

```
}
```

```
}
```

6. Write a Java program to update a student's marks based on the student ID using the Statement interface.

```
import java.sql.*;
```

```
public class Q6 {
```

```
    public static void main(String args[]) {
```

```
        try {
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            Connection con = DriverManager.getConnection(
```

```
                "jdbc:mysql://localhost:3306/studentdb",
```

```
                "root",
```

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```
        "root"  
    );  
  
    Statement st = con.createStatement();  
  
    st.executeUpdate("update Student set marks=90 where id=101");  
  
    System.out.println("Record Updated");  
  
    con.close();  
} catch (Exception e) {  
    System.out.println(e);  
}  
}  
}
```

7. Write a Java program to delete a student record using the Statement interface

```
import java.sql.*;
```

```
public class Q7 {  
    public static void main(String args[]) {  
        try {  
            Class.forName("com.mysql.cj.jdbc.Driver");  
  
            Connection con = DriverManager.getConnection(  
                "jdbc:mysql://localhost:3306/studentdb",  
                "root",  
                "root"  
            );  
  
            Statement st = con.createStatement();
```

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```
st.executeUpdate("delete from Student where id=101");
```

```
System.out.println("Record Deleted");
```

```
con.close();
```

```
} catch (Exception e) {
```

```
    System.out.println(e);
```

```
}
```

```
}
```

```
}
```

8. Write a Java program to insert employee details using the PreparedStatement interface.

```
import java.sql.*;
```

```
public class Q8 {
```

```
    public static void main(String args[]) {
```

```
        try {
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            Connection con = DriverManager.getConnection(
```

```
                "jdbc:mysql://localhost:3306/employeedb",
```

```
                "root",
```

```
                "root"
```

```
            );
```

```
            PreparedStatement ps = con.prepareStatement(
```

```
                "insert into Employee values(?,?,?)"
```

```
            );
```

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```
ps.setInt(1, 1);
```

```
ps.setString(2, "Ravi");
```

```
ps.setInt(3, 25000);
```

```
ps.executeUpdate();
```

```
System.out.println("Employee Inserted");
```

```
con.close();
```

```
} catch (Exception e) {
```

```
    System.out.println(e);
```

```
}
```

```
}
```

```
}
```

9. Write a Java program to update an employee's salary using PreparedStatement.

```
import java.sql.*;
```

```
public class Q9 {
```

```
    public static void main(String args[]) {
```

```
        try {
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            Connection con = DriverManager.getConnection(
```

```
                "jdbc:mysql://localhost:3306/employeeedb",
```

```
                "root",
```

```
                "root"
```

```
            );
```

```
            PreparedStatement ps = con.prepareStatement(
```

```
                "update Employee set salary=? where id=?"
```

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```
);
```

```
ps.setInt(1, 30000);
```

```
ps.setInt(2, 1);
```

```
ps.executeUpdate();
```

```
System.out.println("Salary Updated");
```

```
con.close();
```

```
} catch (Exception e) {
```

```
    System.out.println(e);
```

```
}
```

```
}
```

```
}
```

10. Write a Java program to delete an employee record using PreparedStatement

```
import java.sql.*;
```

```
public class Q10 {
```

```
    public static void main(String args[]) {
```

```
        try {
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
            Connection con = DriverManager.getConnection(
```

```
                "jdbc:mysql://localhost:3306/employeedb",
```

```
                "root",
```

```
                "root"
```

```
            );
```

```
            PreparedStatement ps = con.prepareStatement(
```

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```
"delete from Employee where id=?"
);

ps.setInt(1, 1);

ps.executeUpdate();

System.out.println("Employee Deleted");

con.close();
} catch (Exception e) {
    System.out.println(e);
}
}
}
```

11. Write a Java program to search for an employee by employee ID using PreparedStatement

```
import java.sql.*;

public class Q11 {
    public static void main(String args[]) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/employeeedb",
                "root",
                "root"
            );
```

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```
PreparedStatement ps = con.prepareStatement(
    "select * from Employee where id=?"
);

ps.setInt(1, 1);

ResultSet rs = ps.executeQuery();

while(rs.next()) {
    System.out.println(rs.getInt(1)+" "+rs.getString(2)+" "+rs.getInt(3));
}

con.close();
} catch(Exception e) {
    System.out.println(e);
}
}
}
```

12. Write a Java program to insert three employee records into the Employee table using the PreparedStatement interface

```
import java.sql.*;
```

```
public class Q12 {
    public static void main(String args[]) {
        try {
```

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```
Class.forName("com.mysql.cj.jdbc.Driver");
```

```
Connection con = DriverManager.getConnection(  
    "jdbc:mysql://localhost:3306/employeedb",  
    "root",  
    "root"  
);
```

```
PreparedStatement ps = con.prepareStatement(  
    "insert into Employee values(?, ?, ?)"  
);
```

```
ps.setInt(1, 101);  
ps.setString(2, "Ravi");  
ps.setInt(3, 25000);  
ps.executeUpdate();
```

```
ps.setInt(1, 102);  
ps.setString(2, "Sita");  
ps.setInt(3, 30000);  
ps.executeUpdate();
```

```
ps.setInt(1, 103);  
ps.setString(2, "Kiran");  
ps.setInt(3, 35000);  
ps.executeUpdate();
```

```
System.out.println("3 Records Inserted");
```

```
con.close();
```

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```
    } catch(Exception e) {  
        System.out.println(e);  
    }  
}  
}
```

13. Create a stored procedure to insert employee details and invoke it using the CallableStatement interface.

```
import java.sql.*;
```

```
public class Q13 {  
    public static void main(String args[]) {  
        try {  
            Class.forName("com.mysql.cj.jdbc.Driver");  
  
            Connection con = DriverManager.getConnection(  
                "jdbc:mysql://localhost:3306/employeedb",  
                "root",  
                "root"  
            );  
  
            CallableStatement cs = con.prepareCall("{call insertEmp(?,?,?)}");  
  
            cs.setInt(1,104);  
            cs.setString(2,"Ramesh");  
            cs.setInt(3,28000);
```

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```
cs.execute();

System.out.println("Record Inserted");

con.close();
} catch(Exception e) {
    System.out.println(e);
}
}
}
```

14. Create a stored procedure to update an employee's salary and execute it using CallableStatement.

```
import java.sql.*;

public class Q14 {
    public static void main(String args[]) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/employeeedb",
                "root",
                "root"
            );
```

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```
CallableStatement cs = con.prepareCall("{call updateSalary(?,?)");
```

```
cs.setInt(1,104);
```

```
cs.setInt(2,40000);
```

```
cs.execute();
```

```
System.out.println("Salary Updated");
```

```
con.close();
```

```
} catch(Exception e) {
```

```
    System.out.println(e);
```

```
}
```

```
}
```

```
}
```

15. Write a Java program to retrieve employee details using a stored procedure and CallableStatement.

```
import java.sql.*;
```

```
public class Q15 {
```

```
    public static void main(String args[]) {
```

```
        try {
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
```

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```
Connection con = DriverManager.getConnection(
    "jdbc:mysql://localhost:3306/employeedb",
    "root",
    "root"
);

CallableStatement cs = con.prepareCall("{call displayEmp()}");

ResultSet rs = cs.executeQuery();

while(rs.next()) {
    System.out.println(rs.getInt(1)+" "+rs.getString(2)+" "+rs.getInt(3));
}

con.close();
} catch(Exception e) {
    System.out.println(e);
}
}
```

16. Write a Java program to demonstrate forward traversal of records using ResultSet

```
import java.sql.*;

public class Q16 {
    public static void main(String args[]) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
```

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```
"jdbc:mysql://localhost:3306/employeeedb",
"root",
"root"
);

Statement st = con.createStatement();

ResultSet rs = st.executeQuery("select * from Employee");

while(rs.next()) {
    System.out.println(rs.getInt(1)+" "+rs.getString(2)+" "+rs.getInt(3));
}

con.close();
} catch(Exception e) {
    System.out.println(e);
}
}
}
```

17. Write a Java program to demonstrate scrollable ResultSet operations using first(), last(), previous(), next(), and absolute() methods

```
import java.sql.*;

public class Q17 {
    public static void main(String args[]) {
```

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```
try {  
    Class.forName("com.mysql.cj.jdbc.Driver");  
  
    Connection con = DriverManager.getConnection(  
        "jdbc:mysql://localhost:3306/employeedb",  
        "root",  
        "root"  
    );  
  
    Statement st = con.createStatement(  
        ResultSet.TYPE_SCROLL_INSENSITIVE,  
        ResultSet.CONCUR_READ_ONLY  
    );  
  
    ResultSet rs = st.executeQuery("select * from Employee");  
  
    rs.first();  
    System.out.println(rs.getString(2));  
  
    rs.last();  
    System.out.println(rs.getString(2));  
  
    rs.previous();  
    System.out.println(rs.getString(2));  
  
    rs.absolute(2);  
    System.out.println(rs.getString(2));  
  
    con.close();  
} catch (Exception e) {
```

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```
        System.out.println(e);
    }
}
}
```

18. Write a Java program to update records using an updatable ResultSet. Program:

```
import java.sql.*;

public class Q18 {
    public static void main(String args[]) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/employeedb",
                "root",
                "root"
            );

            Statement st = con.createStatement(
                ResultSet.TYPE_SCROLL_INSENSITIVE,
                ResultSet.CONCUR_UPDATABLE
            );

            ResultSet rs = st.executeQuery("select * from Employee");
```

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```
        if(rs.next()) {
            rs.updateInt("salary",50000);
            rs.updateRow();
        }

        System.out.println("Record Updated");

        con.close();
    } catch(Exception e) {
        System.out.println(e);
    }
}
}
```

19. Write a Java program to insert multiple employee records using JDBC Batch Updates.

```
import java.sql.*;

public class Q19 {
    public static void main(String args[]) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/employeedb",
                "root",
```

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```
"root"  
);  
  
Statement st = con.createStatement();  
  
st.addBatch("insert into Employee values(201,'Ram',25000)");  
st.addBatch("insert into Employee values(202,'Hari',28000)");  
st.addBatch("insert into Employee values(203,'Kiran',30000)");  
  
st.executeBatch();  
  
System.out.println("Batch Insert Completed");  
  
con.close();  
} catch(Exception e) {  
    System.out.println(e);  
}  
}  
}
```

20. Write a Java program to update multiple employee records using Batch Processing.

```
import java.sql.*;  
  
public class Q20 {  
    public static void main(String args[]) {
```

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```
try {  
    Class.forName("com.mysql.cj.jdbc.Driver");  
  
    Connection con = DriverManager.getConnection(  
        "jdbc:mysql://localhost:3306/employeedb",  
        "root",  
        "root"  
    );  
  
    Statement st = con.createStatement();  
  
    st.addBatch("update Employee set salary=35000 where id=201");  
    st.addBatch("update Employee set salary=36000 where id=202");  
  
    st.executeBatch();  
  
    System.out.println("Batch Update Completed");  
  
    con.close();  
} catch (Exception e) {  
    System.out.println(e);  
}  
}
```

21. Write a Java program to delete multiple records using Batch Updates.

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```
import java.sql.*;

public class Q21 {
    public static void main(String args[]) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/employeedb",
                "root",
                "root"
            );

            Statement st = con.createStatement();

            st.addBatch("delete from Employee where id=201");
            st.addBatch("delete from Employee where id=202");

            st.executeBatch();

            System.out.println("Batch Delete Completed");

            con.close();
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```